Markov Chains
Summer of Science Plan of Action

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Roll Number: 25B3009
Mentor: Abhineet Majety

Duration: 8 Weeks

Weekly Timeline

Week 1 (Day 1–7)

Objective: Build foundational understanding of probability

Theory:

- Probability spaces, axioms, conditional probability, Bayes' Theorem
- Independence and combinatorial reasoning
- Distributions: Bernoulli, Binomial, Poisson, Uniform, Normal
- Expectation, variance, covariance

Resources:

- Ross, *Introduction to Probability and Statistics* (Ch. 3)
- Hoel, Port, Stone, *Introduction to Probability Theory* (Sec. 1.1–1.5)

CS229 Lectures:

- Lectures 1–3

Problems:

- Fifty Challenging Problems: 1–6

Week 2 (Day 8–14)

Objective: Random variables and Markov property

Theory:

- Random variables (discrete and continuous)
- Expectation, variance, distributions
- Introduction to stochastic processes
- Markov property and memorylessness
- Transition diagrams and Chapman–Kolmogorov equations

Resources:

- Ross (Ch. 4–5)
- Hoel–Port–Stone (Ch. 3–4)

CS229 Lectures:

- Lectures 4–6

Problems:

- Fifty Challenging Problems: 7–13

Week 3 (Day 15–21)

Objective: Discrete-Time Markov Chains

Theory:

- Definition of DTMCs and transition matrices
- Multi-step transitions and matrix powers
- State classification: transient, recurrent, communicating
- Stationary distributions and limiting behavior
- Ergodic intuition

Resources:

- Norris, *Markov Chains* (Sec. 1.1)
- Ross, *Stochastic Processes* (intro sections)

CS229 Lectures:

- Lectures 7–9

Problems:

- Fifty Challenging Problems: 14–20

Week 4 (Day 22–28)

Objective: Continuous-time processes and Poisson models

Theory:

- Continuous-Time Markov Chains (CTMCs)
- Transition rates and Q-matrix intuition
- Poisson process and exponential waiting times
- Memorylessness in continuous time

Resources:

- Norris (CTMC intro)
- Ross, *Stochastic Processes* (Poisson process)

CS229 Lectures:

- Lectures 10–11

Problems:

- Fifty Challenging Problems: 21–26

Midterm Report Submission

Week 5 (Day 29–35)

Objective: Markov Decision Processes

Theory:

- MDPs: states, actions, transitions, rewards
- Policies and expected return
- Value functions and Bellman equations

Resources:

- CS229 Reinforcement Learning Lectures
- Sutton & Barto, *Reinforcement Learning* (Ch. 3–4)

CS229 Lectures:

- Lectures 12–14

Problems:

- Fifty Challenging Problems: 27–33

Week 6 (Day 36–42)

Objective: Reinforcement Learning fundamentals

Theory:

- Value iteration and policy iteration
- Convergence intuition
- Model-free RL introduction

Resources:

- CS229 RL Lectures
- Sutton & Barto (Ch. 4–6)

CS229 Lectures:

- Lectures 15–16

Problems:

- Fifty Challenging Problems: 34–40

Week 7 (Day 43–49)

Objective: RL implementation

Implementation:

- Build Q-learning agent
- Train and analyze convergence
- Plot performance curves

Resources:

- CS229 RL Lectures
- Sutton & Barto (Q-learning sections)

CS229 Lectures:

- Lectures 17–19

Problems:

- Fifty Challenging Problems: 41–45

Week 8 (Day 50–56)

Objective: Revision and consolidation

Work:

- Revise probability and Markov chains
- Refine RL implementation
- Final report preparation

Resources:

- All previous materials
- Sutton & Barto (review sections)

CS229 Lectures:

- Lectures 20–21

Problems:

- Fifty Challenging Problems: 46–56

EndTerm Report Submission

References

- Sheldon M. Ross, *Introduction to Probability and Statistics for Engineers and Scientists*
- Sheldon M. Ross, *Stochastic Processes*
- J. R. Norris, *Markov Chains*
- Hoel, Port, Stone, *Introduction to Probability Theory*
- Frederick Mosteller, *Fifty Challenging Problems in Probability*
- Sutton & Barto, *Reinforcement Learning: An Introduction*
- Stanford CS229: Machine Learning (Andrew Ng Lecture Series)